

Fórmula del núcleo de Fejér

Lema 1. $\forall N \in \mathbb{N} : F_N(x)$ es 1-periódica. Además,

$$\forall N \in \mathbb{N}, x \in [-1/2, 1/2) : F_N(x) = \begin{cases} \frac{1}{N} \left(\frac{\sin(\pi Nx)}{\sin(\pi x)} \right)^2 & x \neq 0, \\ N & x = 0. \end{cases}$$

Demostración: En primer lugar, para $x = 0$, tenemos que

$$F_N(0) = \frac{1}{N} \sum_{k=0}^{N-1} D_k(0) \stackrel{\text{lem-formula-nucleo-dirichlet/1}}{=} \frac{1}{N} \sum_{k=0}^{N-1} (2k+1) = \frac{1}{N} [1 + 3 + 5 + \dots + (2N-3) + (2N-1)] = N.$$

Si $x \neq 0$, por el `lem-formula-nucleo-dirichlet`/Lema 1, tenemos que

$$F_N(x) = \frac{1}{N} \sum_{k=0}^{N-1} \frac{\sin((2k+1)\pi x)}{\sin(\pi x)}.$$

Aplicando que $2(\sin(\pi x))(\sin((2k+1)\pi x)) = \cos(2k\pi x) - \cos(2(k+1)\pi x)$ obtenemos

$$\begin{aligned} F_N(x) &= \frac{1}{2N} \frac{1}{\sin^2(\pi x)} \sum_{k=0}^{N-1} [\cos(2k\pi x) - \cos(2(k+1)\pi x)] \\ &= \frac{1}{N} \frac{1}{\sin^2(\pi x)} \frac{1 - \cos(2N\pi x)}{2} = \frac{1}{N} \left(\frac{\sin(\pi Nx)}{\sin(\pi x)} \right)^2. \end{aligned}$$

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